

Programming Project 2

*Instructor: Kelin Luo***Deadline: Nov/17/2025**

Your Name: _____ Your Student ID: _____

Problems	1	2	3	4	Total
Max. Score	10	10	10	10	40
Your Score					

Submission requirement:

- You should submit **each source file for each problem**. Name your file: Problem number_Last name_First name_YourStudents ID Number_ProjectNumber.
Example: 1_Doe_John_55552222_P2
- Allowed programming languages: Python. Use Python version ≥ 3.4 . You can run “python -version” to check the version. Please do not use any built-in functions unless they are explicitly stated in the provided code. Do not use any third-party packages other than the standard library.
- Please follow the correct input and output format of P1 to complete P2, and create a separate file for each problem.
- The auto similarity-check will be applied for the programming submissions.
- The P1 deadline is 17 Nov, 11:59PM EST. Late submissions (within 24 hours with 25 % penalty or 48 hours with a 50% penalty) will be recorded according to the submission timestamp on UBlerns. Submissions will close 48 hours after the deadline.
- Only the most recent submission is kept on Brightspace. Note that in each submission, please submit **all** your source files. Only files with extension .py will be accepted for the submission. You should view your submission after you upload it to make sure that it is not corrupted or malformed. You are responsible for making sure your submission went through successfully.
- The grading scheme and test cases will be released when the grades for Programming Project 2 are published.

Problem 1 (10 points). You need to implement the divide and conquer algorithm to compute $a^n \bmod m$. An $O(\log n)$ -time algorithm is sufficient to pass any feasible test cases.

- **Input** You need to read the input from the console. The input consists of one line of data. There are three positive integers a , n and m . You can assume that $1 \leq a \leq 10$, $1 \leq n \leq 10000$ and $2 \leq m \leq 10$.
- **Output** You need to output to console. The output is the number $a^n \bmod m$.

Example: Below are examples of input and output:

Example input: 2 5 3	Example output: 2
-------------------------	----------------------

Example input: 3 1000 6	Example output: 3
----------------------------	----------------------

Problem 2 (10 points). Background: You are given two DNA sequences, `sequence1` with size n characters and `sequence2` with m characters. DNA sequences are composed of the characters 'A', 'C', 'G', and 'T'. Each DNA sequence may contain two consecutive sequences: `sequence1=sequence11+sequence12` and `sequence2=sequence21+sequence22`. For example, `sequence1='AGACTACCG'`, `sequence1` may contain two consecutive sequences `sequence11=''` and `sequence12='AGACTACCG'`, or `sequence1` may contain two sequences `sequence11='A'` and `sequence12='GACTACCG'`, ... or `sequence1` may contain two sequences `sequence11='AGACTACCG'` and `sequence12=''`.

Your task is to find two sequences, `sequence12` and `sequence22`, such that: (1) The length of the longest common subsequence (LCS) of `sequence12` and `sequence22` is equal to the length of the LCS of `sequence1` and `sequence2`. (2) The total number of characters in `sequence12` and `sequence22` is minimized. The output should be the length of the LCS and the total length of the two sequences, `sequence12` and `sequence22`, that you have found.

An $O(nm)$ -time algorithm is suffice to pass any feasible test cases.

Hint: use the dynamic programming algorithm for longest common subsequence problem.

- (a) **Input** You need to read the input from the console. It contains two lines, each containing one sequence. In the first line of the input, we have n characters; and in the second line of the input, we have m characters. You can assume that $1 \leq n \leq 10000$ and $1 \leq m \leq 10000$.
- (b) **Output** You need to output to the console. The first line is an integer indicating the length of the LCS between the given two sequences. The second line is the total length of the two sequences, `sequence12` and `sequence22`.

Below are example for input and output:

Example input: AGACTTAA ATGACTATTCA	Example output: 7 19
Example input: ACTTAA TGCATTCA	Example output: 4 11

Problem 3 (10 points). Given are an undirected graph with n nodes (vertices) and m edges, each having weights. Your task is to find a minimum spanning forest containing no more than k trees. Note that $1 \leq k \leq n$.

An $O(m \log^2 n)$ -time algorithm is sufficient to pass all test cases.

Hint: use Kruskal's algorithm.

- (a) **Input** You need to read the input from the console. In the first line of the input, we have three positive integers n , m and k . n is the number of vertices in the graph, m is the number of edges in the graph and k is a positive number. The vertices are indexed from 1 to n . You can assume that $1 \leq n \leq 1000$, $1 \leq m \leq 100000$ and $1 \leq k \leq n$. In the next m lines, each line contains 3 integers: u , v and w , with $1 \leq u < v \leq n$ and $1 \leq w \leq 100000$. This indicates that there is an edge (u, v) of weight $w > 0$. You can also assume that the graph is connected.
- (b) **Output** You need to output to the console. The output is an integer indicating the total weight of the minimum spanning forest that contains no more than k trees, i.e., the minimum spanning forest that contains $\leq k$ trees.

Below are example for input and output:

Example input:	Example output:
5 5 1 1 2 2 1 3 1 2 3 2 2 4 3 4 5 4	10

Example input:	Example output:
5 5 2 1 2 2 1 3 1 2 3 2 2 4 3 4 5 4	6

Problem 4 (10 points). Given are a directed graph with n nodes (vertices) and m edges, each having positive weights. Your task is to find the shortest path from vertex 1 to vertex 2.

An $O(n^2)$ -time algorithm is sufficient to pass all test cases.

- (a) **Input** You need to read the input from the console. In the first line of the input, we have two positive integers n and m . n is the number of vertices in the graph and m is the number of edges in the graph. The vertices are indexed from 1 to n . You can assume that $2 \leq n \leq 1000$ and $1 \leq m \leq 5000$. In the next m lines, each line contains 3 integers: u , v and w , with $1 \leq u \leq n$, $1 \leq v \leq n$ and $0 \leq w \leq 1000$. This indicates that there is an edge (u, v) from vertex u to vertex v .
- (b) **Output** You need to output to the console. The output is an integer indicating the total length of the shortest path from vertex 1 and vertex 2. If there is no path from vertex 1 to vertex 2 in the graph, the output is INFINITY.

Below are example for input and output:

Example input:	Example output:
5 5 1 2 5 1 3 1 2 5 2 2 4 3 3 2 2	3

Example input:	Example output:
5 4 1 5 2 1 3 4 2 3 2 5 4 1	INFINITY